MIL-F-45133B (AR)
AMENDMENT 6
30 August 1982
SUPERSEDING
AMENDMENT 5
1 June 1974

MILITARY SPECIFICATION

FIRE CONTROL COMPONENTS AND ASSEMBLIES

This amendment forms a part of Military Specification MIL-F-45133B(Ord), dated 18 January 1962, and is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

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* 2.1, Specifications, Military, Delete reference to MIL-I-45208 in its entirety.

Add the following references:

MIL-S-6872	Soldering Process, General Specification for
MIL-S-45743	Soldering, Manual Type, High Reliability, Electrical Connections, for Missile Systems, Procedures for
MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type
* MIL-B-48612	Bonding with Epoxy-Elastomeric Adhesive System, Glass to Metal

* 2.1, Standards, Military:

Delete reference to MIL-STD-10 in its entirety.

Add: MIL-STD-130 Identification Marking of U.S. Military Property

* Add: DOD-STD-100 Engineering Drawing Practice

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* 2.2, Other Publications:

Change reference to Handbook H28 to read "FED-STD-H28"

FSC-12GP

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* Add the following: "AMERICAN NATIONAL STANDARDS INSTITUTE
ANSI B46.1-1978 Surface Texture, Surface Roughness, Waiviness and Lay
ANSI Y14.5-1973 Dimensioning and Tolerancing

(Application for copies shall be addressed to the American National Standards Institute, 1430 Broadway, New York City, New York 10018)."

3.3, line 1: Insert the following as the first sentence: "Interpretation of tolerance symbols specified on drawings shall be as defined in ANSI Y14.5 - 1973."

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* 3.4, Retain Table I. Delete paragraph and substitute:

"3.4 Machine finish. Where letter symbols for surface roughness of machined finishes are specified on drawings, roughness height as defined in American Standard ANSI B46.1-1978 shall be in accordance with Table I. The specified finish value represents the roughest surface acceptable, regardless of the material specified or the process used to obtain the finish."

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3.7.2, line 1: Delete "800 volts" and substitute "500 volts".

3.8, Delete and substitute:

"3.8 Part identification and marking requirements. Part identification and marking requirements shall be in accordance with MIL-STD-130, Identification Marking of U.S. Military Property."

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3.9.4, Delete and substitute:

"3.9.4 Soldering.

3.9.4.1 Soldering of electrical connections. Soldering of electrical connections and certification of soldering operators shall be in accordance with MIL-S-45743.

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3.9.4.2 Soldering, except electrical connections. Soldering, other than electrical connections, shall be in accordance with MIL-S-6872. The solder type designation shall be as specified on the applicable drawing."

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3.9.4.3, Add the following:

"3.9.4.3 Soldering of printed wiring assemblies. Soldering of printed wiring assemblies shall be in accordance with MIL-S-46844, Solder Bath, Soldering of Printed Wiring Assemblies, Automatic Machine Type."

* 3.9.5, Delete and substitute:

3.9.5 Bonded joints. The bonding of optical glass components to their metal supports shall be in accordance with MIL-B-48612. Bonding of other than glass to metal shall be in accordance with the recommended practice of the material manufacturer unless specified techniques are prescribed on the drawings or referenced specifications."

3.10, Add the following:

"3.10 Source control drawings. A source control drawing is one which defines the required item entirely, or in part, by means of a vendor's part number. It may include additional functional, test or configuration specifications but since these are in themselves inadequate to assure proper performance, installation, and interchangeability of the item in a specific application, procurement of the item is thus effectively restricted to the approved source or sources listed on the drawing.

3.10.1 Substitute items. Where source control drawings in accordance with the provisions of DOD-STD-100 (Engineering Drawing Practices) are included in the contractual technical data package for commercial or vendor items, the contractor shall supply only the approved item(s) specified on the drawing unless written approval is obtained from the Contracting Office for use of a substitute item. The contractor is cautioned, however, that the acquisition of data necessary to substantiate the physical and functional interchangeability of a proposed substitute for an approved source control item, and its long-term reliability in its intended environment may require facilities, equipment, and

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resources not available or practically obtainable at the time approval of a substitute item is requested. The Government, therefore, makes no representation as to the feasibility or probability of approving a substitute item. Inasmuch as the contractor is obligated to supply the specified item or an approved equal, rejection or delay in approval by the Government of a proposed substitute item shall not be construed as justification for any change in contract terms and conditions.

3.10.2 Unavailability of specified approved item. Where an item specified on a source control drawing can be documented by the contractor to have been commercially available to him at the time of submitting a bid or proposal, but should thereafter be withdrawn from the commercial market, it shall be the contractor's responsibility to submit evidence of the interchangeability and reliability of an available substitute item. The Government will in such cases assume responsibility for such evaluation and testing as it deems necessary to assure equality of the proposed substitute; however, unless the Government was advised of the unavailability of the specified item prior to contract finalization, delay in proposing and obtaining approval of the substitute item shall not be cause for change in the terms and conditions of the contract.

3.10.3 Suitability of approved item. Unless otherwise specified in the contract, the Government will assume responsibility for the suitability of approved items for their intended use, provided that the items comply with all specified requirements of the source control drawing."

* 4.2: Delete in its entirety.

* 4.3: Delete in its entirety.

* 4.3.1: Delete in its entirety.

* 4.4: Delete in its entirety.

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* 4.4.1: Delete in its entirety.

* 4.4.2: Delete in its entirety.

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* 4.4.3, Add the following subparagraphs:

"4.4.3.1 Contractor design of test equipment. The contractor shall design and supply inspection equipment compatible with the component inspection procedures specified in 4.6.6. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.3.2 Gaging for screw threads. Thread gages utilized for inspection of screw threads shall be made in accordance FED-STD-H28."

* 4.4.4: Delete in its entirety.

* 4.5: Delete in its entirety.

* 4.6: Change to read "Quality Conformance Inspection".

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* 4.6.3: Change "Class 1 Defects" to "Major Defects".

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4.6.6.1.1, Delete and substitute:

"4.6.6.1.1 Alternate procedure. Where a SQAP specifies the use of TABLES II and III in MIL-F-45133, MIL-STD-105 shall be utilized in lieu of the deleted TABLES II and III with the following AQL's applicable except where specific AQL's are designated:

Major Defects - AQL 0.65%
Minor Defects - AQL 2.5%"

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* 4.6.4: Change "Class 2 Defects" to "Minor Defects".

4.6.6.2, Delete and substitute:

"4.6.6.2 Components not covered by SQAP. Components not covered by SQAP shall be inspected in accordance with sampling plan tables of MIL-STD-105 and the AQL's specified in 4.6.6.1.1."

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TABLES II and III: Delete in their entirety.

PAGE 13

* 6.1: Delete in its entirety.

* 6.2: Delete 6.2b and 6.2c in their entirety.

* 6.2.1: Delete in its entirety.

* 6.2.1.1: Delete in its entirety.

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* 6.2.1.2: Delete in its entirety

* 6.2.1.3: Delete in its entirety

* 6.2.1.4: Delete in its entirety.

* 6.2.2: Delete in its entirety.

* The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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